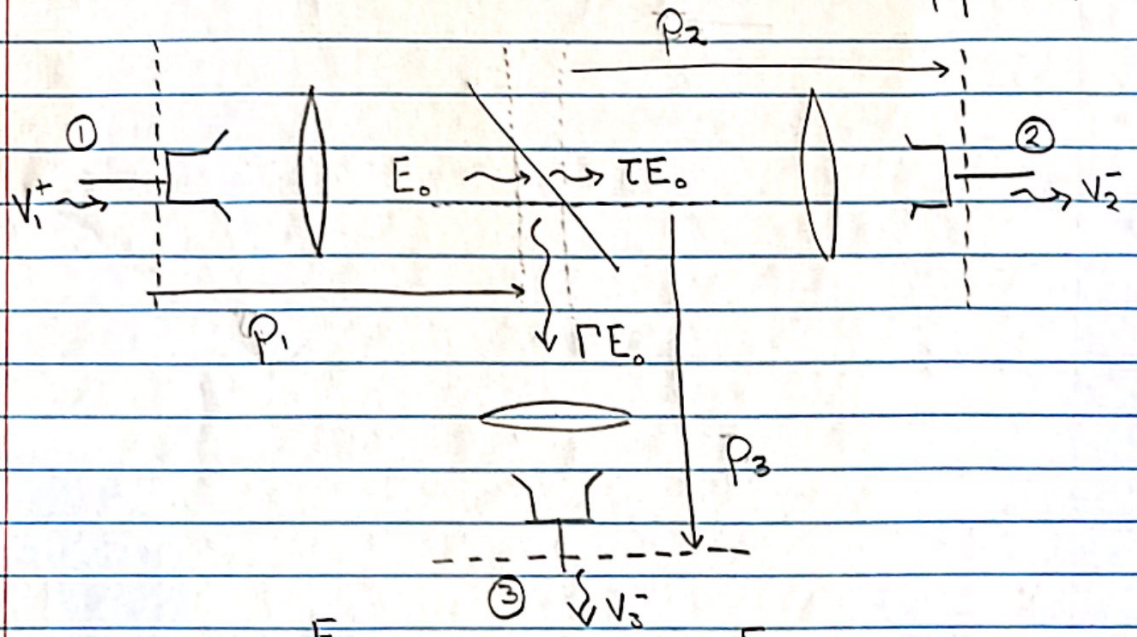
Calibration Procedure (Approx)



$$V_3^- = \rho_3 \Gamma \overbrace{\rho_1 V_1^+}^{E_0}, \quad V_2^- = \rho_2 \tau \overbrace{\rho_1 V_1^+}^{E_0}$$

No material $\rightarrow \Gamma = 0, \tau = 1$

$$V_3^- = 0, \quad V_2^- = \rho_2 E_0$$

Metal sheet (0 thickness) $\Gamma = -1$

$$V_3^- = -\rho_3 E_0, \quad V_2^- = 0$$

Sheet Impedance (0 thickness)

$$V_3^- = \rho_3 \Gamma_s E_0, \quad V_2^- = \rho_2 \tau_s E_0$$

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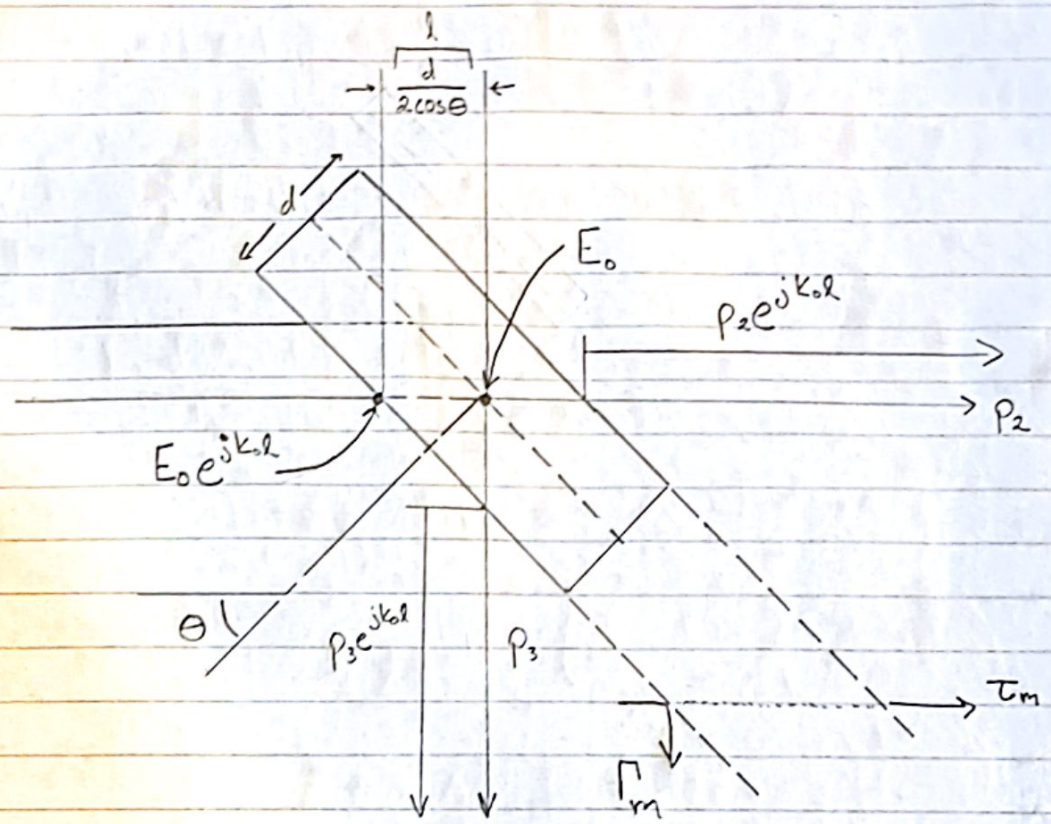
C2



$$V_3^- = E_0 e^{jk_0 l} \cdot \Gamma_m \cdot \rho_3 e^{jk_0 l} \quad , \quad V_2^- = E_0 e^{jk_0 l} \cdot \tau_m \cdot \rho_2 e^{jk_0 l}$$

$$V_3^- = \Gamma_m E_0 \rho_3 e^{j2k_0 l} \quad , \quad V_2^- = \tau_m E_0 \rho_2 e^{j2k_0 l}$$

Through Measurement $V_2^- \triangleq T(\theta) = \rho_2 E_0$

$$V_2^- = \tau_m T e^{j2k_0 l}$$

Reflect Measurement $V_3^- = -E_0 \rho_3 e^{jk_0 d_{sheet}/\cos\theta} = G(\theta)$

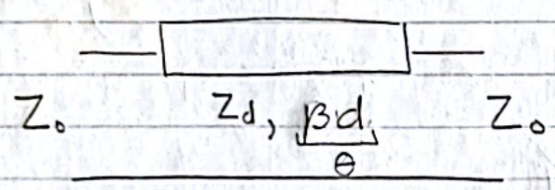
$$E_0 \rho_3 = -G(\theta) e^{-jk_0 d_{sheet}/\cos\theta}$$

$$V_3^- = -\Gamma_m G(\theta) \exp\left(j \frac{k_0 (d - d_{sheet})}{\cos\theta}\right)$$

$$\tau_m = \frac{V_2^-}{T} \exp\left[-j\frac{k_0 d}{\cos\theta}\right]$$

$$\Gamma_m = \frac{-V_3^-}{G} \exp\left[-j\frac{k_0(d-d_{sheet})}{\cos\theta}\right]$$

Dielectric Example



$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \cos\beta d, & jZ_d \sin\beta d \\ j\frac{1}{Z_d}\sin\beta d, & \cos\beta d \end{bmatrix}$$

$$S_{11} = \frac{\cos\beta d + j\frac{Z_d}{Z_0}\sin\beta d - j\frac{Z_0}{Z_d}\sin\beta d - \cos\beta d}{2\cos\beta d + j\left(\frac{Z_d}{Z_0}+\frac{Z_0}{Z_d}\right)\sin\beta d}$$

$$\Gamma_m^{exp} = \frac{j\left(\frac{Z_d}{Z_0}-\frac{Z_0}{Z_d}\right)\sin\beta d}{2\cos\beta d + j\left(\frac{Z_d}{Z_0}+\frac{Z_0}{Z_d}\right)\sin\beta d}$$

$$Z_{TE}^d = \frac{\omega\mu}{\beta} = \eta_0 \frac{k_0 d}{\beta d}$$

$$Z_{TE}^0 = \frac{\omega\mu}{\beta_0} = \eta_0 \frac{k_0 d}{\beta_0 d}$$

$$\tau_m^{exp} = S_{21} = \frac{2}{2\cos\beta d + j\left(\frac{Z_d}{Z_0}+\frac{Z_0}{Z_d}\right)\sin\beta d}$$

$$r = \frac{Z_{TE}^d}{Z_{TE}^0} = \frac{\beta_0 d}{\beta d}$$

$$\beta = \sqrt{k_0^2\epsilon_r - k_0^2\sin^2\theta} = k_0\sqrt{\epsilon_r - \sin^2\theta}$$

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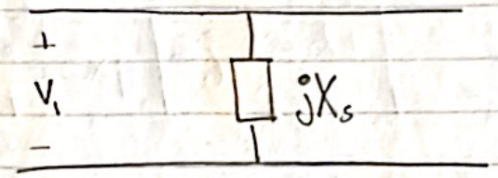

$$Z_{in} = \left(Z_0^{-1} - jX_s^{-1}\right)^{-1} = \frac{Z_0}{1 - j\frac{Z_0}{X_s}}$$

$$S_{11} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{Z_0 - \left(1 - j\frac{Z_0}{X_s}\right)Z_0}{Z_0 + \left(1 - j\frac{Z_0}{X_s}\right)Z_0} = \frac{jZ_0^2/X_s}{2Z_0 - jZ_0^2/X_s}$$

$$y = Z_0/X_s$$

$$= \frac{jy}{2 - jy} \qquad y = \frac{\eta_0/\cos\theta}{X_s} = \frac{1}{\frac{X_s}{\eta_0}\cos\theta}$$

$$S_{21} = 1 + S_{11} = \frac{2 - jy + jy}{2 - jy} = \frac{2}{2 - jy}$$

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